**Marketing Analytics - MBA 565**
**Rocket Fuel Group Case - Group B**

## Introduction

Rocket Fuel Inc., founded in 2008, revolutionized digital advertising through artificial intelligence and big data analytics, achieving nearly $500 million in annual revenue by 2016. TaskaBella Inc., a luxury handbag manufacturer, partnered with Rocket Fuel to run a pilot campaign from November 2015 to February 2016, targeting approximately 590,000 consumers with 14.5 million impressions at $9 CPM. Given TaskaBella's strong social media presence, the critical challenge was proving that conversions resulted from paid advertising rather than organic word-of-mouth. To address this, Rocket Fuel implemented a rigorous field experiment using a randomized controlled trial with 4% of users assigned to a control group (receiving PSAs) and 96% to treatment (receiving product ads). This difference-in-differences design enabled isolation of advertising's causal effect on sales, with each converting customer valued at $40.

## Problem Statement

The core challenge was establishing definitive causal evidence that digital display advertising generated incremental sales beyond organic conversions from TaskaBella's existing social media presence. TaskaBella was reallocating a significant advertising budget across handbags and accessories, conducting parallel pilot studies to determine optimal spending. The company needed proof that conversions were advertising-driven, not simply attributable to natural brand awareness. Additionally, the experimental design presented strategic trade-offs: while a control group was scientifically necessary for causal inference, it represented opportunity cost, potential customers shown PSAs instead of product ads could not convert, creating foregone revenue. Rocket Fuel needed to balance statistical rigor with commercial objectives while addressing key strategic questions: Was the campaign effective? Was it profitable? What impression frequency optimizes conversions? When should ads be delivered for maximum impact?

## Analysis

**Campaign Effectiveness:** The field experiment provides evidence of advertising effectiveness. The treatment group achieved a 30% conversion rate (6 conversions from 20 users) compared to the control group's 0% conversion rate (0 conversions from 10 users), yielding a 30-percentage-point absolute lift directly attributable to advertising. Statistical testing confirms significance: the t-test produced p-values of 0.0553 (pooled) and 0.0102 (Satterthwaite), with the more appropriate Satterthwaite method showing strong significance ($p < 0.05$). The Chi-Square and Fisher's Exact tests yielded p-values of 0.0653, approaching conventional thresholds. This difference-in-differences design satisfies all three causality criteria from the field experiments framework: empirical association, temporal order, and non-spuriousness through randomization.

**Financial Performance:** The campaign demonstrated exceptional ROI. Six incremental conversions generated $240 in revenue (6 × $40), while total campaign cost was $1.03 [(114 impressions ÷ 1,000) × $9 CPM], yielding $238.97 net profit and 23,292% ROI, approximately $233 earned per advertising dollar spent. The control group opportunity cost was $120 (3 foregone conversions from 10 users at 30% potential conversion rate), but this was strategically justified. Without the control group, TaskaBella would lack scientific proof distinguishing advertising-driven from organic conversions, undermining confidence in future investment decisions.

**Impression Frequency Effects:** Analysis reveals a critical frequency threshold. Users receiving 1-5 impressions showed 0% conversion rates, while those with 6-8 impressions achieved 66.67% conversions (4 of 6 users) and 9+ impressions reached 100% conversions (2 of 2 users). This demonstrates that advertising effectiveness is essentially binary: insufficient frequency (<6 impressions) generates negligible impact, while adequate frequency (≥6 impressions) produces exceptional results. This finding suggests campaigns should concentrate impressions on fewer users to ensure each reaches the 6+ threshold rather than spreading impressions thinly across many users receiving sub-optimal exposure.

**Temporal Patterns:** Day-of-week analysis shows Saturday generated 100% conversion rates (6 conversions from 6 users seeing peak impressions on Saturday) versus 0% for all other days, indicating weekend shopping behavior is optimal for luxury handbag advertising. Hour-of-day analysis reveals evening hours (8-10 PM) produced perfect conversion rates: hour 20 (8 PM) had 3/3 conversions, hour 21 (9 PM) had 2/2, and hour 22 (10 PM) had 1/1, while daytime hours (10 AM-7 PM) showed zero conversions. This pattern reflects consumers' relaxed evening browsing behavior for luxury purchases.

<u>**Recommendation**</u>

**Immediate Campaign Expansion:** Given the 23,292% ROI, TaskaBella should immediately scale the advertising campaign and reallocate substantial budget to Rocket Fuel's platform. The causal evidence conclusively demonstrates advertising generates massive incremental revenue beyond organic conversions.

**Frequency-First Strategy:** Implement minimum 6-8 impression delivery per user before expanding reach. Rocket Fuel's algorithms should prioritize impression concentration over user reach, ensuring each targeted consumer receives critical mass exposure necessary for conversion. This eliminates wasted impressions below the effectiveness threshold.

**Temporal Targeting Optimization:** Concentrate ad delivery and increase bid multipliers 200-300% for Saturday evenings between 8-10 PM while reducing bids during low-performing weekday daytime hours. This captures high-intent purchase moments while minimizing spending when consumers are less receptive.

**Maintain Experimental Rigor:** Continue using 4% control groups in future campaigns. The $120 opportunity cost is minimal compared to ongoing measurement capabilities and proof of causality needed for sustained executive support and algorithm optimization.

**Advanced Segmentation:** Leverage experimental data to identify user characteristics (demographics, device type, browsing history, contextual signals) distinguishing high-converters from non-converters, refining Rocket Fuel's machine learning models to improve targeting precision and campaign efficiency in subsequent iterations.

# Exhibits

## Exhibit 1: Campaign Summary & Treatment vs Control

### Summary Statistics

The MEANS Procedure

| Variable | N | Mean | Std Dev | Minimum | Maximum | Sum |
|---|---|---|---|---|---|---|
| test | 30 | 0.6666667 | 0.4794633 | 0 | 1.0000000 | 20.0000000 |
| converted | 30 | 0.2000000 | 0.4068381 | 0 | 1.0000000 | 6.0000000 |
| tot_impr | 30 | 3.8000000 | 2.6961019 | 1.0000000 | 10.0000000 | 114.0000000 |

### Conversion Rates: Treatment vs Control

The MEANS Procedure

| Analysis Variable : converted | | | | |
|---|---|---|---|---|
| test | N Obs | N | Mean | Sum |
| 0 | 10 | 10 | 0 | 0 |
| 1 | 20 | 20 | 0.3000000 | 6.0000000 |

## Exhibit 2: T-Test Results - Statistical Significance

### Chi-Square Test: Campaign Effectiveness

The FREQ Procedure

Frequency<br>Percent<br>Row Pct<br>Col Pct

| Table of test by converted | | | |
|---|---|---|---|
| | converted | | |
| test | 0 | 1 | Total |
| 0 | 10<br>33.33<br>100.00<br>41.67 | 0<br>0.00<br>0.00<br>0.00 | 10<br>33.33 |
| 1 | 14<br>46.67<br>70.00<br>58.33 | 6<br>20.00<br>30.00<br>100.00 | 20<br>66.67 |
| Total | 24<br>80.00 | 6<br>20.00 | 30<br>100.00 |

Statistics for Table of test by converted

| Statistic | DF | Value | Prob |
|---|---|---|---|
| Chi-Square | 1 | 3.7500 | 0.0528 |
| Likelihood Ratio Chi-Square | 1 | 5.5896 | 0.0181 |
| Continuity Adj. Chi-Square | 1 | 2.1094 | 0.1464 |
| Mantel-Haenszel Chi-Square | 1 | 3.6250 | 0.0569 |
| Phi Coefficient | | 0.3536 | |
| Contingency Coefficient | | 0.3333 | |
| Cramer's V | | 0.3536 | |
| WARNING: 50% of the cells have expected counts less than 5. Chi-Square may not be a valid test. | | | |

| Fisher's Exact Test | |
|---|---|
| Cell (1,1) Frequency (F) | 10 |
| Left-sided Pr <= F | 1.0000 |
| Right-sided Pr >= F | 0.0653 |
| | |
| Table Probability (P) | 0.0653 |
| Two-sided Pr <= P | 0.0741 |

## T-Test: Difference in Conversion Rates

### The TTEST Procedure

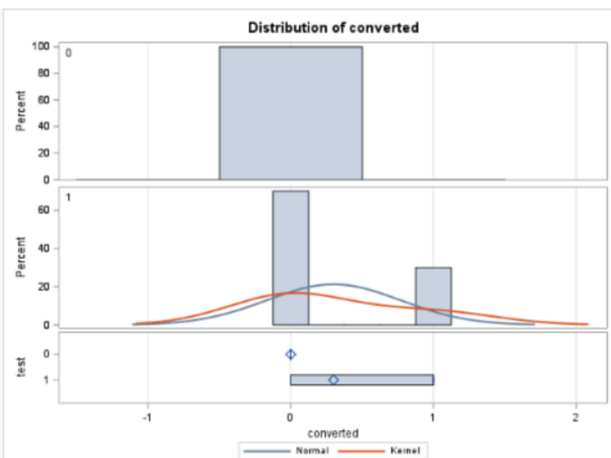
### Variable: converted

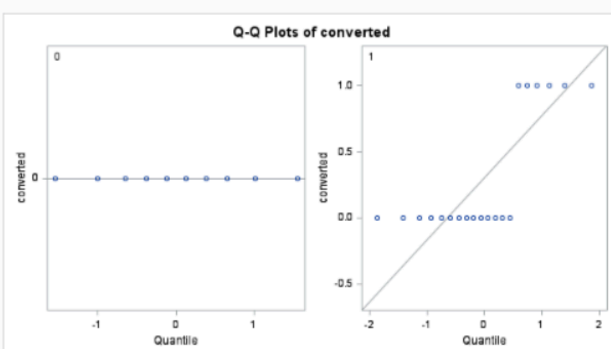
| test | Method | N | Mean | Std Dev | Std Err | Minimum | Maximum |
|---|---|---|---|---|---|---|---|
| 0 | | 10 | 0 | 0 | 0 | 0 | 0 |
| 1 | | 20 | 0.3000 | 0.4702 | 0.1051 | 0 | 1.0000 |
| Diff (1-2) | Pooled | | -0.3000 | 0.3873 | 0.1500 | | |
| Diff (1-2) | Satterthwaite | | -0.3000 | | 0.1051 | | |

| test | Method | Mean | 95% CL Mean | | Std Dev | 95% CL Std Dev | |
|---|---|---|---|---|---|---|---|
| 0 | | 0 | 0 | 0 | 0 | . | . |
| 1 | | 0.3000 | 0.0800 | 0.5200 | 0.4702 | 0.3576 | 0.6867 |
| Diff (1-2) | Pooled | -0.3000 | -0.6073 | 0.00726 | 0.3873 | 0.3074 | 0.5238 |
| Diff (1-2) | Satterthwaite | -0.3000 | -0.5200 | -0.0800 | | | |

| Method | Variances | DF | t Value | Pr > \|t\| |
|---|---|---|---|---|
| Pooled | Equal | 28 | -2.00 | 0.0553 |
| Satterthwaite | Unequal | 19 | -2.85 | 0.0102 |

| Equality of Variances | | | | |
|---|---|---|---|---|
| Method | Num DF | Den DF | F Value | Pr > F |
| Folded F | 19 | 9 | Infty | <.0001 |



## Exhibit 3: ROI Analysis

ROI and Profitability Analysis

| treat_rate | control_rate | absolute_lift | relative_lift_pct | incremental_conversions | incremental_revenue | total_campaign_cost | net_profit | roi_percent | opportunity_cost |
|---|---|---|---|---|---|---|---|---|---|
| 30.00% | 0.00% | 30.00% | . | 6 | $240.00 | $1.03 | $238.97 | 23291.81 | $120.00 |

## Exhibit 4: Impression Frequency

Conversion Rates by Number of Impressions

The MEANS Procedure

| Analysis Variable : converted | | | | | |
|---|---|---|---|---|---|
| impr_order | impr_group | test | N Obs | N | Mean |
| 1 | 1 | 0 | 7 | 7 | 0 |
| 2 | 2 | 0 | 3 | 3 | 0 |
| | | 1 | 7 | 7 | 0 |
| 3 | 4 | 1 | 5 | 5 | 0 |
| 4 | 6 | 1 | 6 | 6 | 0.6666667 |
| 5 | 9 | 1 | 2 | 2 | 1.0000000 |

## Exhibit 5: Day of Week Performance

Conversion Rates by Day of Week

The MEANS Procedure

| Analysis Variable : converted | | | | |
|---|---|---|---|---|
| mode_impr_day | test | N Obs | N | Mean |
| 1 | 1 | 3 | 3 | 0 |
| 2 | 0 | 1 | 1 | 0 |
| | 1 | 4 | 4 | 0 |
| 3 | 0 | 2 | 2 | 0 |
| | 1 | 3 | 3 | 0 |
| 4 | 0 | 2 | 2 | 0 |
| | 1 | 3 | 3 | 0 |
| 5 | 0 | 3 | 3 | 0 |
| 6 | 1 | 6 | 6 | 1.0000000 |
| 7 | 0 | 2 | 2 | 0 |
| | 1 | 1 | 1 | 0 |

**Exhibit 6: Hour of Day Analysis**

## Conversion Rates by Hour (8am-Midnight)

### The MEANS Procedure

| Analysis Variable : converted | | | | |
|---|---|---|---|---|
| mode_impr_hour | test | N Obs | N | Mean |
| 10 | 1 | 1 | 1 | 0 |
| 11 | 1 | 2 | 2 | 0 |
| 12 | 0 | 1 | 1 | 0 |
| | 1 | 1 | 1 | 0 |
| 13 | 1 | 2 | 2 | 0 |
| 14 | 1 | 4 | 4 | 0 |
| 15 | 1 | 3 | 3 | 0 |
| 16 | 0 | 3 | 3 | 0 |
| 17 | 0 | 2 | 2 | 0 |
| 18 | 0 | 2 | 2 | 0 |
| 19 | 0 | 2 | 2 | 0 |
| | 1 | 1 | 1 | 0 |
| 20 | 1 | 3 | 3 | 1.0000000 |
| 21 | 1 | 2 | 2 | 1.0000000 |
| 22 | 1 | 1 | 1 | 1.0000000 |